KALAINESAN NATARAJAN

Computer Science Undergraduate | Backend Systems | AI Agents | Machine Learning

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EDUCATION

SRM Institute of Science and Technology, Kattankulathur, India *Expected Graduation: 2029*
B.Tech in Computer Science and Engineering

GEMS Our Own High School, Al Warqa'a, Dubai *2013 - 2023*
Primary and Secondary School Education

TECHNICAL SKILLS

Programming:	Python, JavaScript, TypeScript, Java, SQL
Backend Engineering:	Node.js, Express.js, REST APIs, Socket.io, PostgreSQL, MongoDB
Frontend Development:	React, Next.js, TypeScript, Tailwind CSS
AI & Machine Learning:	LLM Agents, Multi-Agent Systems, Prompt Engineering, Scikit-learn, Pandas, NumPy
Tools & Infrastructure:	Git, GitHub, Docker, Supabase, MongoDB Atlas, Vercel, Render

FEATURED PROJECTS

BullWarden / TriAgent Advisory System *[Multi-Agent AI | Financial Technology (OSSome Hacks 3.0)]*
- Built autonomous multi-agent AI trading architecture with Analyst, Risk, and Trader agents using OpenClaw.
- Enforced ArmorClaw / ArmorIQ runtime security policies with prompt injection defense & Alpaca paper trading API.
- Executed LLM-based financial news sentiment analysis with structured audit logging and strict risk classification.

Offline Sync Conflict Resolution Engine *[Backend Engineering | Distributed Systems (GDG on Campus SRM)]*
- Engineered backend synchronization engine handling concurrent offline edits with optimistic concurrency control.
- Implemented PostgreSQL SELECT FOR UPDATE, field-level conflict detection/merging, and idempotency guarantees.
- Documented endpoints with Swagger/OpenAPI and validated integrity with comprehensive Jest/Supertest suite.

QR-Based Geo-Tagged Attendance System *[Full-Stack Development | Web Application (NSCC Technical Club)]*
- Developed full-stack verification system pairing dynamic QR codes with GPS geofencing via Haversine formula.
- Created real-time organizer dashboards using Socket.io, JWT auth, MongoDB Atlas, and CSV export capabilities.

CampusBorrow - Peer-to-Peer Campus Equipment Sharing *[Full-Stack Development | Next.js 16 & Supabase]*
- Architected student lending platform with Next.js 16, React 19, TypeScript, and Supabase Row Level Security.
- Enforced atomic PostgreSQL RPC row-locking for race-condition prevention during simultaneous item claims.

Air Quality Forecasting Pipeline *[Machine Learning | Data Science (AI/ML Club Recruitment)]*
- Formulated 1-hour ahead CO concentration predictor using Linear Regression and engineered temporal lag features.
- Evaluated across 5-fold TimeSeriesSplit yielding R2: 0.7377, RMSE: 0.6939, MAE: 0.4652 (10.7% MAE improvement).

TECHNICAL ACTIVITIES & CLUB RECRUITMENT

- OSSome Hacks 3.0: Built TriAgent financial advisory system with multi-agent runtime safety policies.
- GDG on Campus SRM: Designed offline synchronization engine with PostgreSQL transactions & idempotency.
- NSCC Technical Club: Implemented GPS-geofenced QR attendance management system.
- AI/ML Club Recruitment: Engineered temporal regression pipeline for ambient air pollution forecasting.